Homework 3

2026-03-27

Question 1

Suppose that airplane engines operate independently and fail with probability equal to 0.4. Assuming that a plane makes a safe flight if at least one-half of its engines run, determine whether a 4-engine plane or a 2-engine plane has the higher probability for a successful flight. Compare probabilities.

Solution:

We use the binomial distribution because this experiment follows the Bernoulli process. For a 4-engine plane, the probability that at least half of engines run is

$$P(X \geq 2) = 1 - P(X \leq 1).$$

For a 2-engine plane, the probability that at least half of engines run is

$$P(Y \geq 1) = 1 - P(Y = 0).$$

```r
p=0.6 # probability of engine successfully working

eng4_success = 1-pbinom(1,4,p)
eng4_success
```

```
## [1] 0.8208
```

```r
eng2_success = 1-dbinom(0,2,p)
eng2_success
```

```
## [1] 0.84
```

```r
eng4_success > eng2_success
```

```
## [1] FALSE
```

Therefore, the 2-engine plane has a higher probability for a successful flight.

Question 2

An electronic switching device occasionally malfunctions, but the device is considered satisfactory if it makes, on average, no more than 0.20 errors per hour. A particular 5-hour period is chosen for testing the device. If no more than 1 error occurs during the time period, the device will be considered satisfactory.

1. What is the probability that a satisfactory device with $\lambda = 1$ will be considered unsatisfactory on the basis of the test?
2. What is the probability that a device will be accepted as satisfactory when, in fact, the mean number of errors per hour is 0.25?

Solution:

The number of errors follows a Poisson distribution. Over 5 hours, the expected number of errors is λt.

Part 1: A satisfactory device has $\lambda = 0.20$ errors/hour, so over 5 hours $\lambda t = 1$. The device is unsatisfactory if more than 1 error occurs: $P(X > 1) = 1 - P(X \leq 1)$.

Part 2: If $\lambda = 0.25$, then over 5 hours $\lambda t = 1.25$. The device is accepted if at most 1 error: $P(X \leq 1)$.

```
# Part 1: lambda*t = 0.20*5 = 1
p_unsatisfactory = 1 - ppois(1, lambda = 1)
p_unsatisfactory
```

```
## [1] 0.2642411
```

```
# Part 2: lambda*t = 0.25*5 = 1.25
p_accepted = ppois(1, lambda = 1.25)
p_accepted
```

```
## [1] 0.6446358
```

Question 3

Find the probability of being dealt a hand of 13 cards containing 5 spades, 2 hearts, 3 diamonds, and 3 clubs.

Solution:

This is a multivariate hypergeometric distribution. We choose 5 from 13 spades, 2 from 13 hearts, 3 from 13 diamonds, and 3 from 13 clubs, out of all ways to choose 13 from 52 cards.

$$P = \frac{\binom{13}{5}\binom{13}{2}\binom{13}{3}\binom{13}{3}}{\binom{52}{13}}$$

```
prob = choose(13,5) * choose(13,2) * choose(13,3) * choose(13,3) / choose(52,13)
prob
```

```
## [1] 0.01293071
```

Question 4:

A traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from Virginia. What is the probability that fewer than 4 of the next 9 vehicles are from out of state?

Solution:

If 75% are from Virginia, then 25% are from out of state. Let X be the number of out-of-state vehicles. $X \sim \text{Binomial}(n = 9, p = 0.25)$. We want $P(X < 4) = P(X \leq 3)$.

```
pbinom(3, 9, 0.25)
```

```
## [1] 0.8342743
```

Question 5

An electronics firm claims that the proportion of defective units from a certain process is 5%. A buyer has a standard procedure of inspecting 15 units selected randomly from a large lot. On a particular occasion, the buyer found 5 defective items.

1. What is the probability of this occurrence given that the claim of 5% defective is correct?

2. Is this occurrence more likely if 60% of the units are defective? Compare probabilities.
3. Assuming the firm's claim is true, what is the probability that the third lot that the buyer inspects is the first lot to have fewer than 3 defective units?
4. Assuming the firm's claim is true, what is the probability that the buyer requires less than 8 lots to find 2 lots with fewer than 3 defective units?

Solution:

Let $X \sim \text{Binomial}(n = 15, p)$ be the number of defective units in a sample.

Part 1: $P(X = 5)$ with $p = 0.05$.

Part 2: $P(X = 5)$ with $p = 0.60$. Compare.

Part 3: Let $q = P(X < 3)$ with $p = 0.05$. The number of lots until the first lot with fewer than 3 defectives follows a geometric distribution. We want $P(Y = 3)$ where $Y \sim \text{Geometric}(q)$.

Part 4: The number of lots until 2 lots with fewer than 3 defectives follows a negative binomial. We want $P(Y < 8)$ where $Y \sim \text{NegBin}(k = 2, q)$.

```r
# Part 1
p1 = dbinom(5, 15, 0.05)
p1
```

```
## [1] 0.0005618772
```

```r
# Part 2
p2 = dbinom(5, 15, 0.60)
p2
```

```
## [1] 0.02448564
```

```r
p2 > p1
```

```
## [1] TRUE
```

```r
# Part 3: probability a lot has fewer than 3 defectives
q = pbinom(2, 15, 0.05)
q
```

```
## [1] 0.9637998
```

```r
# P(third lot is first with < 3 defectives) = geometric with success prob q
# P(Y=3) = (1-q)^2 * q
p3 = dgeom(3-1, q)
p3
```

```
## [1] 0.001263018
```

```r
# Part 4: P(fewer than 8 lots to find 2 lots with < 3 defectives)
# Y ~ NegBin(k=2, q), P(Y < 8) = P(Y <= 7)
p4 = pnbinom(7-2, 2, q)
p4
```

```
## [1] 1
```

Question 6

A certain area of the eastern United States is, on average, hit by 6 hurricanes a year. Find the probability that, in a given year, that area will be hit by

1. fewer than 4 hurricanes.

2. anywhere from 6 to 8 hurricanes.

Solution:

Let $X \sim \text{Poisson}(\lambda = 6)$.

```
# Part 1: P(X < 4) = P(X <= 3)
ppois(3, 6)
```

```
## [1] 0.1512039
```

```
# Part 2: P(6 <= X <= 8) = P(X <= 8) - P(X <= 5)
ppois(8, 6) - ppois(5, 6)
```

```
## [1] 0.4015579
```

Question 7

It is estimated that 4000 of the next 10000 voting residents of Roanoke are against a new sales tax. If 15 eligible voters are selected at random and asked their opinion, what is the probability that at most 7 favor the new tax?

Solution:

6000 out of 10000 favor the tax, and 4000 are against. This is sampling without replacement, so we use the hypergeometric distribution. Let X be the number who favor the tax. $X \sim \text{Hypergeometric}(N = 10000, K = 6000, n = 15)$. We want $P(X \leq 7)$.

```
# phyper(x, m, n, k) where m=successes in pop, n=failures in pop, k=sample size
phyper(7, 6000, 4000, 15)
```

```
## [1] 0.2129377
```

Question 8

A company purchases large lots of a certain kind of electronic device. A method is used that rejects a lot if 2 or more defective units are found in a random sample of 100 units.

1. What is the mean number of defective units found in a sample of 100 units if the lot is 1% defective?
2. What is the variance?
3. What is the probability of rejecting 2 of the next 3 lots?

Solution:

Let $X \sim \text{Binomial}(n = 100, p = 0.01)$ be the number of defective units in a sample.

Parts 1 & 2: Mean $= np$, Variance $= np(1-p)$.

Part 3: The probability of rejecting a lot is $P(X \geq 2) = 1 - P(X \leq 1)$. The number of rejected lots out of 3 follows a binomial distribution.

```
n = 100; p = 0.01

# Part 1: Mean
mu = n * p
mu
```

```
## [1] 1
```

```r
# Part 2: Variance
variance = n * p * (1 - p)
variance
```

```
## [1] 0.99
```

```r
# Part 3: P(reject a lot) = P(X >= 2)
p_reject = 1 - pbinom(1, n, p)
p_reject
```

```
## [1] 0.264238
```

```r
# P(reject exactly 2 of 3 lots)
dbinom(2, 3, p_reject)
```

```
## [1] 0.1541165
```

Question 9

Suppose X follows a continuous uniform distribution from 1 to 5. Determine the conditional probability $P(X > 2.5 \mid X \leq 4)$.

Solution:

By definition of conditional probability:

$$P(X > 2.5 \mid X \leq 4) = \frac{P(2.5 < X \leq 4)}{P(X \leq 4)}$$

```r
# P(2.5 < X <= 4)
p_num = punif(4, 1, 5) - punif(2.5, 1, 5)

# P(X <= 4)
p_den = punif(4, 1, 5)

# Conditional probability
p_num / p_den
```

```
## [1] 0.5
```

Question 10

Find the value of z if the area under a standard normal curve

1. to the right of z is 0.3622
2. to the left of z is 0.1131
3. between 0 and z (with $z > 0$) is 0.4838
4. between $-z$ and z (with $z > 0$) is 0.95

Solution:

```r
# Part 1: P(Z > z) = 0.3622, so P(Z <= z) = 1 - 0.3622
qnorm(1 - 0.3622)
```

```
## [1] 0.3525844
```

```r
# Part 2: P(Z < z) = 0.1131
qnorm(0.1131)
```

```
## [1] -1.210206
```

```r
# Part 3: P(0 < Z < z) = 0.4838, so P(Z < z) = 0.5 + 0.4838
qnorm(0.5 + 0.4838)
```

```
## [1] 2.139441
```

```r
# Part 4: P(-z < Z < z) = 0.95, so P(Z < z) = (1 + 0.95)/2 = 0.975
qnorm(0.975)
```

```
## [1] 1.959964
```

Question 11

A fair coin is tossed 400 times. Use the normal approximation to find the probability of obtaining

1. between 185 and 210 heads inclusive.
2. exactly 205 heads.
3. fewer than 176 or more than 227 heads.

Solution:

$X \sim \text{Binomial}(n = 400, p = 0.5)$. We approximate with $N(\mu = np = 200, \sigma = \sqrt{np(1-p)} = 10)$, using continuity correction.

```r
mu = 400 * 0.5
sigma = sqrt(400 * 0.5 * 0.5)

# Part 1: P(185 <= X <= 210) approx P(184.5 < Z < 210.5)
pnorm(210.5, mu, sigma) - pnorm(184.5, mu, sigma)
```

```
## [1] 0.7925702
```

```r
# Part 2: P(X = 205) approx P(204.5 < Z < 205.5)
pnorm(205.5, mu, sigma) - pnorm(204.5, mu, sigma)
```

```
## [1] 0.03519553
```

```r
# Part 3: P(X < 176 or X > 227) approx P(Z < 175.5) + P(Z > 227.5)
pnorm(175.5, mu, sigma) + (1 - pnorm(227.5, mu, sigma))
```

```
## [1] 0.01012257
```

Question 12

In Blacksburg, the daily consumption of water (in thousands of liters) approximately follows a gamma distribution with $\alpha = 2$ and $\beta = 3$.

1. If the daily capacity of Blacksburg is 9 thousand liters of water, what is the probability that on any given day the water supply is inadequate?
2. According to Chebyshev's theorem, there is a probability of at least 3/4 that the water consumption on any given day will fall within what interval?

Solution:

$X \sim \text{Gamma}(\alpha = 2, \beta = 3)$. Mean $= \alpha\beta = 6$, Variance $= \alpha\beta^2 = 18$, so $\sigma = \sqrt{18}$.

Part 1: $P(X > 9) = 1 - P(X \leq 9)$.

Part 2: By Chebyshev's theorem, $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$. For probability $\geq 3/4$, we need $1 - 1/k^2 = 3/4$, so $k = 2$. The interval is $\mu \pm 2\sigma$.

```r
alpha = 2; beta = 3

# Part 1: P(X > 9). In R, pgamma uses rate = 1/beta
1 - pgamma(9, shape = alpha, rate = 1/beta)
```

```
## [1] 0.1991483
```

```r
# Part 2: Chebyshev interval with k=2
mu = alpha * beta
sigma = sqrt(alpha * beta^2)
c(mu - 2*sigma, mu + 2*sigma)
```

```
## [1] -2.485281 14.485281
```

Question 13

The life (in years) of a certain type of electrical switch has an exponential distribution with an average life $\beta = 2$. If 100 of these switches are installed in different systems, what is the probability that at most 30 fail during the first year? Solve directly, then use a normal approximation. Discuss the approximation.

Solution:

First, find the probability a single switch fails within the first year: $P(T \leq 1)$ where $T \sim \text{Exp}(\beta = 2)$.

Then the number of switches that fail, $X \sim \text{Binomial}(n = 100, p)$. We want $P(X \leq 30)$.

```r
# Probability a single switch fails in first year
p = pexp(1, rate = 1/2)
p
```

```
## [1] 0.3934693
```

```r
# Direct: P(X <= 30) using binomial
direct = pbinom(30, 100, p)
direct
```

```
## [1] 0.03347124
```

```r
# Normal approximation with continuity correction
mu = 100 * p
sigma = sqrt(100 * p * (1-p))
approx = pnorm(30.5, mu, sigma)
approx
```

```
## [1] 0.03507284
```

```r
cat("Direct:", direct, "\nApprox:", approx, "\nDifference:", abs(direct - approx))
```

```
## Direct: 0.03347124
## Approx: 0.03507284
## Difference: 0.001601607
```

The normal approximation is very close to the exact binomial probability because np and $n(1-p)$ are both well above 5, satisfying the conditions for the approximation.

Question 14

Average power usage (dB per hour) for a particular company is studied and is known to have a lognormal distribution with parameters $\mu = 4$ and $\sigma = 2$. What is the probability that the company uses more than 270 dB during any particular hour?

Solution:

$X \sim \text{LogNormal}(\mu = 4, \sigma = 2)$. We want $P(X > 270) = 1 - P(X \leq 270)$.

```r
1 - plnorm(270, meanlog = 4, sdlog = 2)
```

```
## [1] 0.212084
```